

## 디지털 신호처리(공) (EE401-01)

|      |                                                |
|------|------------------------------------------------|
| 학기   | 2026년 1학기                                      |
| 담당교수 | 이민선                                            |
| 이수구분 | 교선평수                                           |
| 학점   | 3.0                                            |
| 강의시간 | 화09:00-10:30(E7 - G01), 목09:00-10:30(E7 - G01) |
| 강의실  |                                                |
| 소속   | 기초학부                                           |

### 교과개요

This course introduces modern digital signal processing (DSP) techniques used across a wide range of applications. Topics include the mathematical foundations of discrete-time signal analysis, the theory and efficient implementation of fast Fourier transform (FFT) algorithms, and the design and realization of digital filters. The course concludes with case studies demonstrating how DSP methods are applied to solve practical, real-world problems.

### 교과목표

This course aims to build a strong foundation in modern digital signal processing (DSP) theory and implementation, and to develop the ability to apply DSP techniques to real-world problems. By the end of the course, students will be able to: Understand the mathematical fundamentals of discrete-time signals and systems (e.g., sampling, convolution, and frequency-domain analysis). Explain the principles of the DFT/FFT and implement efficient FFT-based spectral analysis. Design and implement digital filters (FIR/IIR) to meet given specifications (e.g., cutoff frequency, ripple, and attenuation). Select, apply, and interpret DSP methods for practical tasks such as noise reduction, feature extraction, and system analysis using real data.

### 주차별 강의계획

- 1주차: Intro to course, review of continuous-time signal/Discrete-time signals and systems
- 2주차: Review of discrete-time signals and systems; convolution/Discrete-time Fourier transform
- 3주차: z-transform/The inverse z-transform
- 4주차: z-transform properties/The discrete Fourier series
- 5주차: The discrete Fourier transform/Sampling of continuous-time signals
- 6주차: Sampling of continuous-time signals/Circular convolution(1)
- 7주차: Midterm
- 8주차: Circular Convolution (2)/Representation of linear digital networks
- 9주차: Network structures for IIR Systems/ Network structures for FIR systems/Parameter quantization effects in digital filter structures
- 10주차: Design of IIR digital filters
- 11주차: Digital Butterworth filters
- 12주차: Design of FIR digital filters
- 13주차: Computation of the discrete Fourier transform
- 14주차: Discrete Hilbert transforms
- 15주차: Final exam
- 16주차:

준비물 및 기타

Course Material